SANT LONGOWAL INSTITUTE OF ENGINEERING AND TECHNOLOGY (Deemed-to-be University under MoE, Govt of India)
Longowal, Punjab – 148106

techFEST '26
SLIET Longowal's Annual Technical Fest

PATRON-IN-CHIEF
Prof. Mani Kant Paswan
Director, SLIET

PATRON
Prof. Vijay Kumar Kukreja
Dean (SW), SLIET

CHAIRMAN
Dr. Sunil Kumar Associate Professor (ME)

VICE-CHAIRMAN
Dr. Barasha Mali AP, (EIE)

VICE-CHAIRMAN
Dr. Tajinder Singh
AP, (CSE)

The Aqua Epoch

Dataset Link :-Aqua Epoch Dataset

ROUND 01 - Report Submission

- PDF submission detailing Data Preprocessing, EDA, Model Architecture and System Flow.
- Round 1 is the mandatory technical evaluation stage for Aqua-Epoch. Teams must submit a detailed project report to demonstrate their ML solution, system architecture and technical approach. Submitting this report is strictly mandatory; missing this step will result in immediate disqualification, and the team will not advance to the next round.
- PDF Submission Checklist
  Submit a single PDF containing:

1. Title Page: Event Name (Aqua-Epoch), Project Title, Team Name/Members, Date, Contact Info.
2. Problem Statement: Brief summary of the core challenge, objectives and predictive tasks.
3. Assumptions: Key assumptions on data availability, feature usage and system design.
4. ML Tasks Description: Framing and targets for Classification (water safety prediction) and Regression (pH prediction).
5. Approach: Preprocessing and feature engineering, model selection/ training methodology, and an overview of system architecture (frontend, backend and inference flow).
6. Results & Insights: Performance summary for both tasks, feature importance, key observations and limitations.
7. Appendix (Optional): Extra plots, dashboards, system diagrams or references.

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ROUND 02 :- Presentation and Demo

- Shortlisted teams demonstrate their web application, with real-time predictions and visualizations, on their own laptops. The application must include:
- Frontend: a simple input form for water metrics (Turbidity, Hardness, TDS) with tooltips on safe limits
- Input validation: values must fall within realistic physical ranges before they are sent to the backend
- Backend API: a REST API that processes the inputs and returns real-time model predictions
- Database (bonus): optional logging of past predictions for trend analysis and tracking.

Key Requirements:

- Presentation: Problem statement, preprocessing, EDA, algorithms, evaluation metrics, and results.
- Live Demo: Demonstrate water potability classification, pH prediction, and visualizations.
- Technical Discussion: Explain model selection, workflow, challenges, limitations, and improvements.

Note: Models must be built from scratch with core machine learning libraries such as Scikit-Learn or XGBoost. Pre-built Generative AI or LLM APIs must not be used for core predictions.
The system is scored on latency and UI quality.

How to Submit:-

- Create a public GitHub repository named **aqua-epoch-<team-name>**
- Upload your code, trained model, web application, requirements.txt and README.md (team name, members, how to run).
- When the Google Form link is shared in the groups, open it and submit the team name, the member details as registered on Unstop, the GitHub repository link.